

Psychometric Evaluation of the Cybersecurity Concept Inventory

Anonymous Author(s)

Abstract

We present a psychometric evaluation of a revised version of the *Cybersecurity Concept Inventory (CCI)*, completed by 355 students from 29 colleges and universities. The CCI is a conceptual test of understanding created to enable research on instruction quality in cybersecurity education. This work extends previous expert review and small-scale pilot testing of the CCI. Results show that the CCI aligns with a curriculum many instructors expect from an introductory cybersecurity course, and that it is a valid and reliable tool for assessing what cybersecurity conceptual knowledge students learned.

CCS Concepts

• **Social and professional topics** → **Student assessment; Computing education**; • **Security and privacy**;

Keywords

Classical Test Theory (CTT), concept inventories, Cybersecurity Assessment Tools (CATS), Cybersecurity Concept Inventory (CCI), cybersecurity education, Item Response Theory (IRT), psychometric evaluation.

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1 Introduction

Knowledge of cybersecurity principles is critical for individuals and organizations to keep sensitive data secure and to avoid theft and other digital threats. Frequent news reports about major security breaches highlight the importance of cybersecurity [34]. There is a large and growing shortage of cybersecurity professionals in the United States [25, 28]. It is essential that we increase the efficiency and effectiveness of educational programs. To conduct reproducible research on the benefits and drawbacks of different teaching methods and curricular structures, validated assessment instruments are needed to measure student knowledge. Until now, no such instrument existed.

After briefly reviewing the creation of the *Cybersecurity Concept Inventory (CCI)*, we give statistical evidence of its reliability and

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Scenario. Alice logs in to a server by sending her username, password, and a timestamp to the server over the Internet. Eve listens to the communications. Eve wishes to log in to Alice's account at a later time.

Question. Choose the modification to the log-in protocol that best deters Eve from later logging in to Alice's account:

- A. Instead of sending the username, password, and timestamp over the Internet, send them in a text message from Alice's smartphone.
- B. In a second step, prompt Alice for additional personal information, such as her mother's maiden name.
- C. Replace the username and password pair with Alice's fingerprint.
- D. Encrypt the username, password, and timestamp.
- E. Include a use-once, randomly generated value to prevent replay attacks.

Figure 1: CCI Question 20 probes the concept “Devise a defense.”

validity as an assessment of cybersecurity knowledge for students who have taken in introductory cybersecurity course.

2 Background

None of the existing cybersecurity assessment instruments (see [10]), seem suitable for use as a research instrument, and we are not aware of any scientific study to characterize psychometric properties of any of them.

2.1 The Cybersecurity Assessment Tools (CATS) Project

The authors founded the *Cybersecurity Assessment Tools (CATS)* project in the interest of creating validated educational assessment tools for cybersecurity [1, 3–10]. Unlike some areas of computer science (CS), in cybersecurity there often is not a clear right or wrong answer to a given problem. Cybersecurity professionals must think deeply about real world scenarios and differentiate what may be poor, mediocre, or ideal solutions to a given security problem. The CCI presents a series of scenarios, and asks questions about the scenario that force students to weigh their options and select the best solution choice to the security problem. Figure 1 gives an example test item from the CCI. In [10], we explain in detail how we created the CCI.

During fall 2018, experts reviewed the CCI and we pilot tested it with a smaller number of students, showing that the CCI assesses topics that cybersecurity educators would expect it to, and that the test had some desirable psychometric properties [9]. Since then, we revised the CCI to improve items that were too hard or did not discriminate well between lower- and higher-performing students. Using a revised version of the test, we started another round of

data collection from fall 2019 through spring 2020. In this paper, we analyze these new data to answer the following research question: “What does the statistical evidence say about the reliability and validity of the CCI?”

2.2 Concept Inventories

A *concept inventory* (CI) is a validated assessment for a given set of topics that enables researchers and instructors to gauge what their students have learned about a given subject. The first CI, the Force Concept Inventory, is credited with helping to realize the active learning revolution in introductory physics by creating a meaningful way to compare the results of different pedagogical techniques [17, 22].

Over the last ten years, computing education researchers have been creating CIs, so our discipline can also benefit from them. Examples include the *Digital Logic Concept Inventory (DLCI)* [21], the *Multilanguage Assessment of CS1 Knowledge (SCS1)* [16, 30, 35], and the *Basic Data Structures Inventory (BDSI)* [31] (for a more extensive review of assessment instruments used in computing education research, see [15, 26]).

The effect of CIs has not yet been as far reaching in CS as it has been in other disciplines, such as physics, likely because CIs have not been in use in CS for as long. Despite their recent creation they have already been useful for many purposes in computing education, including but not limited to: examining the relationship between spatial ability and learning programming [13], comparing outcomes between digital logic courses which use differing pedagogical approaches [19], evaluating novel instructional practices in CS1 [27, 37], evaluating the effectiveness of teaching students using both block- and text-based programming languages [12], and understanding the impact of students’ educational background on learning topics in computer science [11]. This activity confirms the utility of creating concept inventories to the computing education community.

3 Methods

We explain how we collected and analyzed data.

3.1 Data Collection

We pursued multiple avenues for recruiting subjects to take the assessment, including emailing professors who do research in cybersecurity, talking to colleagues, and contacting institutions involved with cybersecurity education programs such as Scholars for Service [14] and institutions qualifying as Centers for Academic Excellence in Cyber Defense (CAEs) [23]. By far the most effective recruitment strategy was making use of the professional connections of the members of our research team, who are embedded in the cybersecurity research and teaching communities [10].

We hosted the CCI in an online, open source homework and exam platform, to facilitate the administration of the assessment to students at a range of institutions. For anonymization purposes [2], we shall call the platform OHEP.

For most students, their instructor offered some extra credit to complete the CCI. We collected data from September 2019 through May 2020. The institutional review board at one of the author’s institution approved our protocol.

A total of 575 students started the CCI in OHEP. After we discarded test instances where the student did not complete the assessment, or spent less than 15 minutes on the assessment, our data set consists of scores from 355 students from 29 colleges and universities (since it takes about 15 minutes just to read all the assessment questions, no student could complete the assessment in good faith in under 15 minutes). In the sanitized data set, the mean time to finish the test is 45 minutes, with 273 out of 355 (77%) test takers finishing in under an hour.

Our participants came from a range of institutions including private and public universities and community colleges. Institutions were geographically diverse within the United States, with some data points also coming from Spain. We collected data from both research-focused and teaching-focused institutions. The data set is biased toward large schools, as they were often able to provide more subjects for testing. Subjects also came from courses with a variety of titles including Computer Security, Cybersecurity Concepts, and Information Assurance and Security, all of which covered most or all of the core cybersecurity topics that the CCI seeks to assess. Most students who took the CCI were CS majors in the latter half of completing their bachelor’s degree.

3.2 Item Response Theory vs. Classical Test Theory

Classical Test Theory (CTT) and *Item Response Theory* (IRT) are two commonly used analytical frameworks for showing statistical support for the validity of assessment instruments, and for gauging the skill of students taking an assessment [24]. Both CTT and IRT give a measurement of each question’s *difficulty*, how hard it is to answer a question correctly, and its *discrimination*, how well a question differentiates between students of lower and higher skill levels. These metrics are defined differently between CTT and IRT, and therefore they should not be compared across frameworks. A robust assessment will have test questions with a range of difficulty levels to obtain information about students at a range of ability levels. It is desirable to have questions with high discrimination, because questions that do a better job differentiating between students of higher or lower ability can measure student ability more accurately.

CTT can be used on samples of any size and is useful for obtaining a simple measurement of the *reliability* of assessment instruments. Some strengths of IRT that CTT does not have are: (1) Falsifiable assumptions: the assumptions of CTT must be taken as a given, whereas the assumptions of IRT can be tested using the data set and appropriate statistical tests. (2) CTT assumes that the measurement error is the same for any student taking the test, where IRT allows us to see if there is a different measurement error for students of different skill levels. (3) IRT enables us to estimate, for each question, how much information the question provides about each student. IRT, while extremely useful, requires a larger sample size than does CTT because it requires fitting a version of the general linear model with a number of parameters which increases with respect to the number of items on the test.

3.3 Classical Test Theory

Our team used CTT to analyze the results of the pilot testing [9]. Here we use CTT as a means of comparing the current properties of the test to the properties of the earlier draft of the test [9], and as a way to verify basic properties of the test, such as whether the CTT difficulty and discrimination fit into the accepted ranges.

CTT assumes that each student has a true score (T) which, together with some error term (E), gives the student's actual score (X), so that $X = T + E$.

3.3.1 Reliability. If a test is *reliable*, a student who takes the same exam repeatedly should receive close to the same score each time they take the test. The most common measure of reliability for assessments is Cronbach's α , because it can be used even if each student has only taken the exam once, as long as the test items are scored dichotomously as correct or incorrect, and under the assumption that the test measures a single latent trait. Cronbach's α which ranges from 0 to 1, with 1 indicating perfect reliability. There is no generally accepted value of Cronbach's α to denote a reliable assessment, but most people agree that for a test to be used in high-stakes scenarios, such as assigning grades, the Cronbach's α should be at least 0.7 or 0.8 [24, 29].

One method for evaluating the quality of items in an assessment is to compare the Cronbach's α with what it would be if the particular item was removed [24]. If removing a test item from the test increases the reliability of the test, the item may be poor quality, and would become a potential candidate for removal.

3.3.2 Difficulty and Discrimination. In CTT, the *difficulty* is the percentage of students who correctly answered a given item, and the *discrimination* is the point biserial correlation between a student's score on the question and their score on the test [21].

3.4 Item Response Theory

We use IRT to gain greater insight into the properties of particular questions, and how much information individual questions and the test as a whole give about students of differing ability levels. We fit the Rasch, *two-parameter logistic (2PL)*, and *three-parameter logistic (3PL)* models using the R package ltm [32, 33]. 3PL did not reach a stable solution. This result is not surprising because it usually takes a large amount of data to fit a model with that many parameters. A likelihood ratio test shows that 2PL does a significantly better job explaining the data than does the Rasch model ($p < 0.001$, LRT = 72.07), so we focus on 2PL.

In 2PL, for each student n and item i , let $X_{n,i} \in \{0, 1\}$ be a dichotomous random variable with $X = 1$ denoting a correct response. We assume that the probability of student n correctly responding to item i can be modeled as a function of the student's ability, θ_n , the difficulty of the item, d_i , and the discrimination of the item a_i as follows:

$$P\{X_{n,i} = 1\} = \frac{1}{1 + e^{-a_i(\theta_n - d_i)}}. \quad (1)$$

The distribution of each student ability parameter θ_n is given a mean of 0 and standard deviation of 1.

3.4.1 Item Response Functions. Inserting the difficulty and discrimination parameters for each test item into Equation 1 gives the

Item	Change in α with item removed	Item	Change in α with item removed
Q1	-0.01	Q14	0.00
Q2	-0.01	Q15	-0.01
Q3	-0.01	Q16	-0.01
Q4	-0.01	Q17	0.00
Q5	-0.01	Q18	0.00
Q6	-0.01	Q19	-0.01
Q7	0.00	Q20	0.00
Q8	-0.01	Q21	-0.01
Q9	-0.01	Q22	-0.01
Q10	-0.01	Q23	-0.01
Q11	-0.01	Q24	0.00
Q12	-0.01	Q25	0.00
Q13	-0.01	All Items	0.78

Table 1: Reliability of the test, and the change in the reliability of the test with each item removed. The overall Cronbach's α is in the acceptable range, and is comparable or better than those of many accepted concept inventories [18, 31, 36]. Removing each item individually results in either the same or lower reliability, suggesting that none of the items on the test are so weak that they should be removed.

item response functions, which help us visualize the difficulty and discrimination of test items.

3.4.2 Item Information Functions. The *item information function* for an item is the derivative of the item response function for that item. It shows how much information that item gives about subjects taking the test. Summing the item information functions for all items on an instrument gives the item information function of the instrument.

Item response theory enables us to use the *standard error of measurement (SE)* for a student based on their ability level:

$$SE(\theta) = \frac{1}{\sqrt{I(\theta)}}, \quad (2)$$

where $I(\theta)$ is the information function of the test.

4 Results

We analyze CCI test data from CTT and IRT perspectives.

4.1 Classical Test Theory Results

CTT reveals information about the CCI's reliability, difficulty, and discrimination.

4.1.1 Reliability. The Cronbach's α of the CCI is 0.78, putting it in the acceptable range for CIs, and comparable to other commonly used CIs in CS, as shown in Table 3. As shown in Table 1, removing each item of the CCI individually results in the same or lower reliability, suggesting that none of the items are so weak that they should be removed.

4.1.2 Difficulty and Discrimination. Figure 2 shows a comparison of the classical test theory difficulty and discrimination of an earlier

Item	Diff. (a_i)	Disc. (d_i)	Item	Diff. (a_i)	Disc. (d_i)
Q1	1.14	0.94	Q14	0.85	0.39
Q2	0.32	1.10	Q15	0.82	1.13
Q3	0.87	0.88	Q16	0.90	0.92
Q4	0.23	1.14	Q17	-0.90	0.65
Q5	0.19	0.80	Q18	-0.94	0.46
Q6	0.34	0.66	Q19	0.04	1.04
Q7	-2.00	0.37	Q20	1.06	0.58
Q8	1.73	0.98	Q21	0.18	0.89
Q9	0.11	1.38	Q22	-0.37	0.95
Q10	-0.63	0.94	Q23	-0.72	1.47
Q11	0.23	0.82	Q24	0.99	0.57
Q12	0.09	0.78	Q25	2.04	0.57
Q13	1.29	1.23			

Table 2: Difficulty and discrimination of each question in the 2PL item response theory model.

version of the CCI (Plot A) compared to those of the current version (Plot B). This figure shows that our revisions to the test successfully strengthened its validity as a measurement of cybersecurity knowledge. This comparison demonstrates the value of continuing to revise and develop an assessment instrument past the initial validation phase.

4.2 Item Response Theory Results

Table 2 shows the difficulty and discrimination parameters for each question as predicted by fitting our data to 2PL.

4.2.1 Item Response Functions. Figure 3 shows the Item Response Functions for the CCI, which represent the probability that a student of a given ability will answer a given item correctly. A steep curve, such as that of Q23, shows that a question has high discrimination, and a more shallow curve, such as that of Q7, shows that a question has lower discrimination. If a question’s item response function has a negative slope (or discrimination), that would mean students with lower ability are more likely to answer the item successfully than are students with higher ability. This outcome would cause serious concern about the usefulness of the test item. None of the questions of the CCI have this problem.

4.2.2 Item Information Functions. Figure 4 shows the item information functions for each of the test items under 2PL. Some test items provide a great deal of information about the student’s ability, such as Q9, while other items provide very little, such as Q7 and the other questions with near-flat item information functions. All the information curves are concave down, showing that none of the test items decrease the amount of information we know about a student’s ability.

Figure 5 shows the item information function for the CCI. The CCI provides the most information about students with $\theta = 0.26$ (students whose ability is 0.26 standard deviations above the mean). The test information curve is greater than 4 on the interval $-0.61 < \theta < 1.17$, telling us that if a student’s ability level is in that range, their ability level can be estimated within ± 0.5 standard deviations with confidence 68%.

5 Discussion

The psychometric evaluation results of the CCI are very promising when compared to the results of other accepted CIs in CS. Table 3 gives a quick overview showing that the statistical properties of the CCI are in the same general range as those for other CIs. In comparison to the SCS1, the CCI does an excellent job providing information about students both above and below mean ability level, whereas the SCS1 is a very difficult test, providing much more information about students above the mean than below [36].

Measurement	CCI	DLCI	SCS1	BDSI
Cronbach’s α	0.78	0.80	0.70	0.68
Min. Difficulty	-2.00	-1.84	0.08	-3.03
Max. Difficulty	2.04	0.55	5.07	1.25
Min. Disc.	0.37	0.28	0.49	0.33
Max. Disc.	1.47	1.68	1.53	2.03

Table 3: Comparison of the reliability and 2PL model parameters of the CCI with those of other CIs. Parameters for other CIs come from [21, 31, 36].

In comparison with the DLCI, however, the CCI has many questions that provide relatively little information, and the test as a whole has more error for students who are far from the mean. We theorize that the nature of cybersecurity questions is such that measuring student ability with low error is more difficult than in some other domains. For example, in the DLCI, most questions have an answer that is clearly correct, and answers that are clearly wrong. On the contrary, many questions on the CCI have answers that are good, better, and best, and we expect a student to pick the “best” answer to be awarded points.

5.1 Expert Analysis of Interesting Items

For many items, we attribute their high discrimination in part to having high-quality distractors—answers that many students pick when they have a common misconception about the cybersecurity scenario. For six items (Q1, Q3, Q8, Q13, Q16, Q20), there was a distractor that subjects found more attractive than the correct answer. We know that the distractor was appealing because the discrimination of the questions were quite high, showing that stronger students were much more likely to answer the question correctly than were weaker students. Removing each of these questions individually from the test decreased the Cronbach’s α of the test as a whole, showing that these questions did help strengthen the reliability of the test.

5.1.1 Question 3. For Question 3 (Figure 6), more students selected Distractor A (122 of 355: 34%) than selected the correct, and somewhat unusual, answer D (120 of 355: 34%). Alternative A is a compelling distractor.

Question 3 probes the important adversarial-thinking concept “Identify the attacker.” The scenario explains why Alice might be motivated to lie by denying having sent the purchase order, which supports D, even though usually Alice would not intentionally reveal her signing key.

Each of the other alternatives can be excluded as implausible. With a strong signature system: signatures cannot be transferred

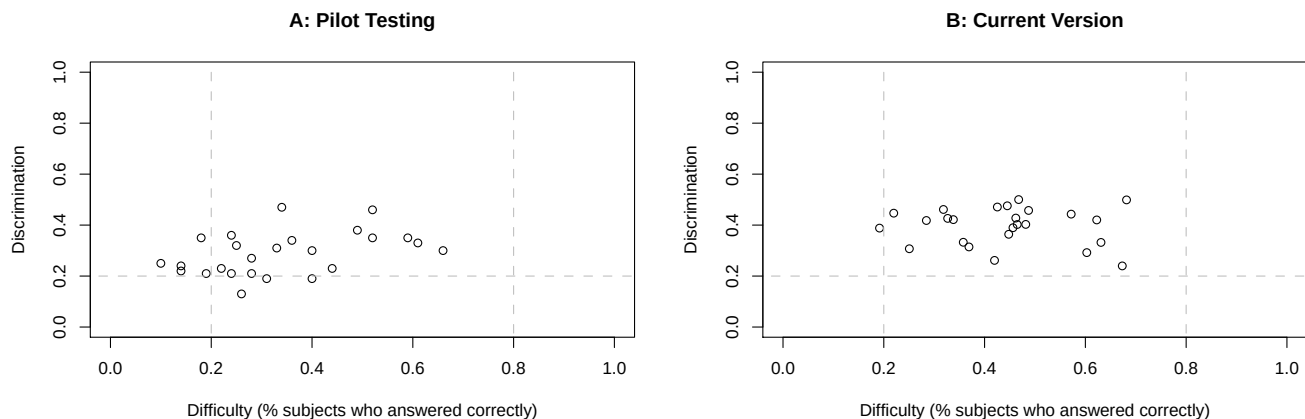


Figure 2: Comparison of the classical test theory difficulty and discrimination of an earlier version of the CCI (Plot A) to that of the current version (Plot B), showing that the revisions to the test successfully strengthened its validity as a measurement of cybersecurity knowledge. All CCI items are now in the accepted discrimination range for Classical Test Theory (above 0.2), and all but one are in the accepted difficulty range (between 0.2 and 0.8) [24]. We created Plot A from data given in [9].

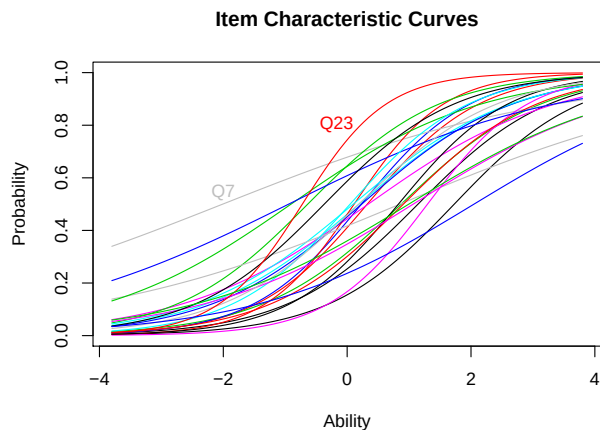


Figure 3: Item characteristic curves from the 2PL IRT model. The differences in slopes of the lines shows the variance of discrimination between parameters. These data confirm that the Rasch model's assumption, that each item has the same discrimination, is not a good fit for our data.

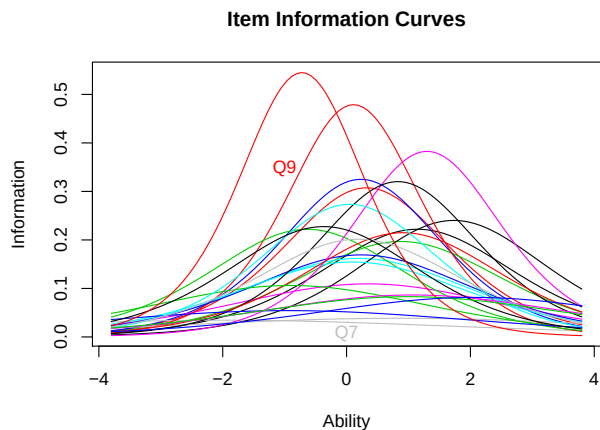


Figure 4: Item information curves from the 2PL IRT model. Some test items provide a great deal of information about the student's ability, while other items provide very little. All the information curves are concave down, showing that none of the test items remove information.

from one document to another (A); the key cannot be deduced from the signature (B); signatures cannot be forged (C); and distinct documents cannot be found that produce identical signatures (E). Distractor A reflects an egregious misconception about digital signatures. Selecting the correct answer requires adversarial thinking and knowledge of fundamental properties of digital signatures. We conjecture that students who picked A may have engaged in improper analogizing, assuming that digital signatures can be copied in the same way that physical signatures can be copied. This conjecture aligns with prior research findings on students using improper analogies while trying to transfer knowledge from one domain to another, as when they assume the properties of an if-then statement in programming and the if-then construct in Boolean logic to be the same [20].

5.1.2 Question 23. Question 23, with four appealing distractors, had the highest discrimination (see Figure 7). The correct answer is Alternative B: to disconnect the local network from the internet. Many students, however, selected Alternative D: protecting the network with a state-of-the-art firewall. This explanation aligns with prior research findings from student interviews that students tend to prefer a digital solution over a physical solution, even in situations where the physical solution does a better job solving the cybersecurity problem [7].

5.2 Using the CCI

We will continue to host the CCI on OHEP [2] for the foreseeable future, and we invite educators to use it and to participate in our ongoing evaluations of it. The authors can forward test results for

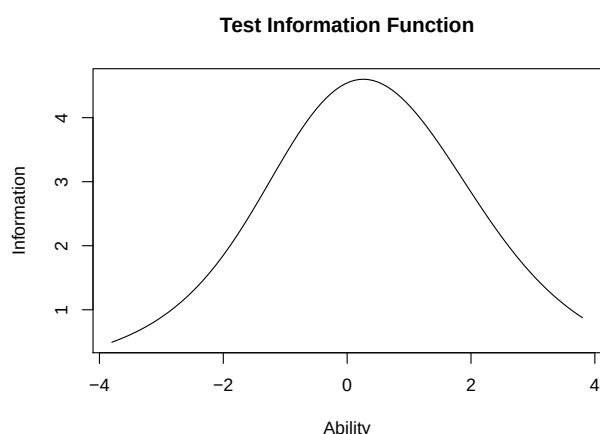


Figure 5: Information curve for the entire test. The CCI gives a reasonable amount of information about students of all ability levels. The slight skew of the test information curve means that the test gives a little bit more information about students who are above average.

Scenario. Alice wants to send a file to Bob over an Internet connection. Bob receives a file digitally signed with Alice's private (signature) key, using a secure digital signature algorithm. The file specifies an electronic order to purchase a large number of shares for a new public offering. Contrary to expectation, the value of the stock plummets. Following this incident, Alice denies having signed the purchase order, pointing out that Charlie has been caught forging her signature.

Question. Choose the most likely explanation for how Charlie forged Alice's signature:

- A. Copied Alice's digital signature from an older electronic purchase order.
- B. Mathematically analyzed Alice's signature to deduce her private key.
- C. Changed bits in Alice's signature to sign another electronic document.
- D. Received Alice's private key from Alice.
- E. Created a new document producing the same digital signature.

Figure 6: CCI Question 3 probes the concept "Identify the attacker." More subjects selected Distractor A than the correct answer.

students who take the CCI through OHEP. The authors are willing to provide a PDF copy, or provide instructions on how someone might host the CCI through OHEP on their own servers.

5.3 Limitations

Most CIs are poorly suited for use as pre-tests, and the CCI has not been administered as a pretest. Therefore, we have no evidence for whether or not it can be use as a reliable pre-test. Given the difficulty of the CCI, this possibility seems unlikely. Also, we are

Scenario. A law firm stores sensitive client records in a database on their local network.

Question. Choose the action that is the MOST likely to prevent an opposing law firm from reading the records:

- A. Require fingerprint scans to access the law offices.
- B. Disconnect their local network from the Internet
- C. Use only trusted vendor software.
- D. Protect the network with a state-of-the-art firewall and intrusion-detection system.
- E. Secure the law offices 24/7 with strong locks and security cameras.

Definitions

24/7: Twenty-four hours a day, seven days a week.

Figure 7: CCI Question 23 probes the concept "Devise a Defense." This question had the highest discrimination of all the questions on the test.

unable to comment on the performance of particular demographic groups, because we did not collect this information.

6 Conclusion

Our psychometric evaluation provides evidence that the CCI is a reliable and valid assessment for classifying the strength of student understanding of basic cybersecurity concepts. Therefore, the CCI can and should be used to compare pedagogic approaches to teaching of cybersecurity.

We plan to apply the CCI to compare the effectiveness of various approaches to teaching and learning cybersecurity. We will also complete our evaluation of a second CI that we developed—the *Cybersecurity Curriculum Assessment (CCA)*, for students completing an undergraduate degree or track in cybersecurity. The CCA targets the same five concepts as does the CCI, but assuming greater technical depth.

CIs are useful tools for promoting change in education through valid and reliable measurement of student knowledge. We have shown that the CCI it is a valid and reliable instrument to measure the cybersecurity knowledge of students who have completed a first course in cybersecurity. We hope that its use will help instructors to diagnose the knowledge of their students, and lead to rigorous research in comparing pedagogic practices in cybersecurity education.

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